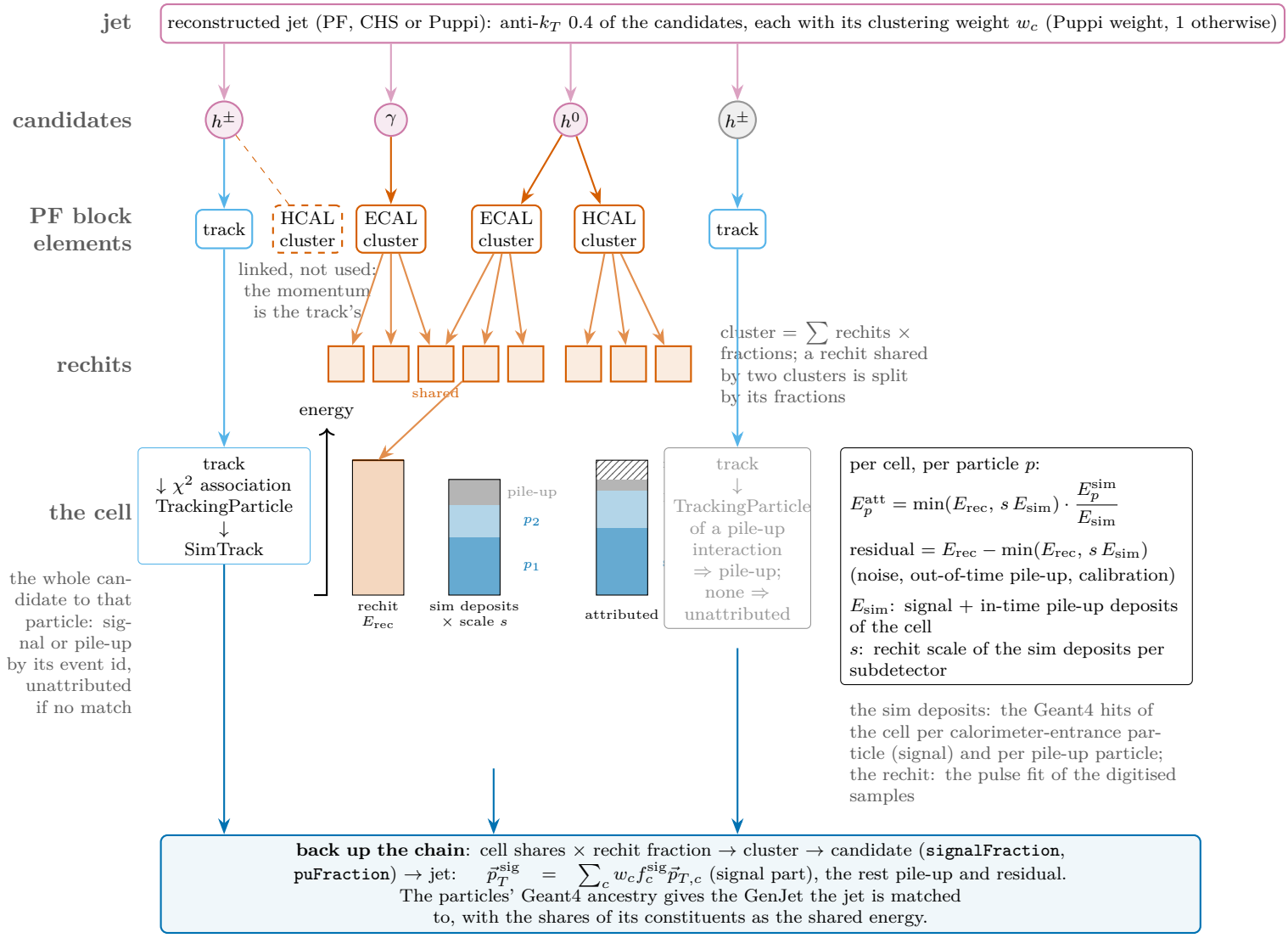

jet
reconstructed jet (PF, CHS or Puppi): anti-k_T 0.4 of the candidates, each with its clustering weight w_c (Puppi weight, 1 otherwise)
candidates
$h^\pm$
γ
h^0
$h^\pm$
PF block elements
track
HCAL cluster
ECAL cluster
ECAL cluster
HCAL cluster
track
linked, not used: the momentum is the track's
rechits
shared
cluster = $\sum$ rechits × fractions; a rechit shared by two clusters is split by its fractions
the cell
energy
track
↓ χ^2 association
TrackingParticle
↓
SimTrack
rechit $E_{\rm rec}$
pile-up
p_2
p_1
sim deposits × scale s
attributed
track
↓
TrackingParticle of a pile-up interaction
⇒ pile-up; none ⇒ unattributed
the whole candidate to that particle: signal or pile-up by its event id, unattributed if no match
per cell, per particle p:
$E_p^{\rm att} = \min(E_{\rm rec},\, s\,E_{\rm sim}) \cdot \frac{E_p^{\rm sim}}{E_{\rm sim}}$
residual = $E_{\rm rec} - \min(E_{\rm rec},\, s\,E_{\rm sim})$
(noise, out-of-time pile-up, calibration)
$E_{\rm sim}$: signal + in-time pile-up deposits of the cell
s: rechit scale of the sim deposits per subdetector
the sim deposits: the Geant4 hits of the cell per calorimeter-entrance particle (signal) and per pile-up particle; the rechit: the pulse fit of the digitised samples
back up the chain: cell shares × rechit fraction → cluster → candidate (signalFraction, puFraction) → jet: $\vec{p}_T^{\,\rm sig} = \sum_c w_c f_c^{\rm sig} \vec{p}_{T,c}$ (signal part), the rest pile-up and residual.
The particles' Geant4 ancestry gives the GenJet the jet is matched to, with the shares of its constituents as the shared energy.